

Introduction to Data Analysis

- Hypothesis testing
- Parametric tests
- Nonparametric tests
- Linear regression



Hypothesis testing

- The **null hypothesis** (H_0) represents a theory that has been presented, either because it is believed to be true or because it is to be used as a basis for an argument.
- It is a statement that has not been proven. It is also important to realize that the null hypothesis is the statement of no difference.
- For example, in a clinical trial of a new drug, the null hypothesis might state that the new drug is no better, on average, than the current drug (in other words, the new drug exhibits the same behavior as the old drug). The **null hypothesis** (H_0) and the **alternative hypothesis** (H_1) can be stated as:
 - H_0 : There is **no difference** between the two drugs.
 - H_1 : There is a significant difference between the two drugs.



- Doing a hypothesis test means making a decision.

	Reject H_0	Retain H_0
H_0 true	Type I error	OK
H_0 false	OK	Type II error



- Mark stands trial for a crime (e.g., burglary).
- What is H_0 , H_1 ? Type I error? Type II error?
- H_0 : Mark is innocent
- H_1 : Mark is guilty
- Type I error: innocent man convicted
- Type II error: guilty man set free



Making the decision on H_0 – p-value

- **p-value** is essentially the significance level
- It shows in how many % of case will you make a mistake if you reject the H_0
- In essence, we are calculating the probability that the H_0 hypothesis is true. It summarizes the credibility of the null hypothesis.

If $p < 0.05$ Reject H_0 in favor of H_1

If $p > 0.05$ Do not reject H_0



Which type of tests to use??

	Parametric	Non-parametric
Assumed distribution	Normal	Any
Assumed variance	Homogeneous	Homogenous and Heterogeneous
Typical data	Ratio or Interval	Ordinal or Nominal
Data set relationships	Independent	Any
Usual central measure	Mean	Median
Benefits	Can draw more conclusions	Simplicity; Less affected by outliers



Which test to use?

Purpose of the test	Parametric Test	Nonparametric Test
Compare two independent samples	Two-sample t-test	Mann-Whitney/ Wilcoxon Rank Sum test
Compare k-independent samples	One-way ANOVA	Kruskal-Wallis test
Compare dependent samples	Paired t-test	Wilcoxon Signed-Rank test



Kolmogorov-Smirnov test

- The Kolmogorov–Smirnov test is a nonparametric test of the equality of continuous, one-dimensional probability distributions that can be used to compare a sample with a reference probability distribution
- In R it needs the parameters of the testing distribution - **Difficulty**
- H_0 : The variable has a particular distribution
- H_1 : The variable does not have a particular distribution



- The alternative for the Kolmogorov–Smirnov test (KS test)
 - Easier to conduct in R
 - More powerful than KS test
 - Test of normality
-
- H_0 : The variable is normally distributed
 - H_1 : The variable is not normally distributed



Shapiro-Wilk test

```
# shapiro-wilk  
idi <- read.csv(file.choose(), header = T)  
shapiro.test(idi$Fixed2012)
```

Value of the
statistics

```
> #Shapiro-wilk test  
> shapiro.test(idi$Fixed2012)  
  
      shapiro-wilk normality test  
  
data:  idi$Fixed2012  
W = 0.90764, p-value = 6.027e-08
```

p value is less than 0.05
Variable Fixed2012 is not normally distributed



Shapiro-Wilk test

```
# Shapiro-wilk  
shapiro.test(idi$Mobile2012)
```

value of the
statistics

```
> shapiro.test(idi$Mobile2012)  
  
shapiro-wilk normality test  
  
data:  idi$Mobile2012  
W = 0.98786, p-value = 0.241
```

p value is greater than 0.05
Variable Mobile2012 is normally distributed



One sample t-test

- The one sample t-test is a statistical procedure used to determine whether a sample of observations could have been generated by a process with a specific mean
- H_0 : The true mean (μ) equals a specific value (μ_0)
- H_1 : The true mean (μ) differs from a specific value (μ_0)



One sample t-test

```
# One sample t-test  
t.test(idi$Mobile2011, mu=95)
```

Testing value of the population mean

value of the
statistics

Mean of
Mobile2011

```
> # One sample t-test  
> t.test(idi$Mobile2011, mu=95)
```

one sample t-test

```
data: idi$Mobile2011  
t = 2.2527, df = 143, p-value = 0.0258  
alternative hypothesis: true mean is not equal to 95  
95 percent confidence interval:  
 95.90076 108.80341  
sample estimates:  
mean of x  
 102.3521  
> |
```

95% confidence
interval of the mean

p value is less than 0.05
The mean of Mobile2011 is not 95



Independent Samples t-test

- An Independent samples t-test is used to determine the likelihood that **two** independent data samples came from populations with identical means.
- If this were true, then the difference between the means should equal zero.
- Two variables are required for the test:
 - One variable is the measured parameter. Examples include weight, height, or frequency. It **must** be normally distributed.
 - The second variable divides the data set into two groups.



Independent Samples t-test

- **Example:** Is there a statistically significant difference in the Mobile subscriptions in 2012 between countries with high HDI index and countries with medium HDI index?
- Pretest is the Levene's test
 - H_0 : variance (high HDI) = variance (medium HDI)
 - H_1 : variance (high HDI) $\neq$ variance (medium HDI)
- T-test
 - H_0 : There is no statistically significant difference between countries with high HDI and countries with medium HDI ($m(\text{high HDI}) = m(\text{medium HDI})$).
 - H_1 : There is statistically significant difference between countries with high HDI and countries with medium HDI ($m(\text{high HDI}) \neq m(\text{medium HDI})$).



Independent Samples t-test

```
# We will make subset for high/medium HDI level  
hmHDI <- subset(idi, (HDI_level=="high" | HDI_level=="medium"))
```

EnvironmentHistoryConnections



Import Dataset



List



Global Environment

Search

Data



hmHDI

66 obs. of 8 variables





idi

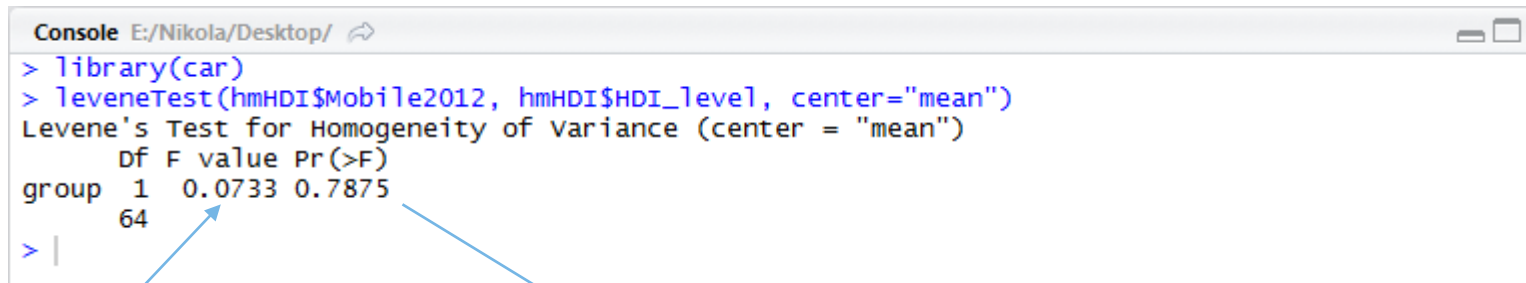
144 obs. of 8 variables





Levene's test

```
# Levene's test for homogeneity of variance
install.packages("car")
library(car)
leveneTest(hmHDI$Mobile2012, hmHDI$HDI_level, center="mean")
```



```
Console E:/Nikola/Desktop/
> library(car)
> leveneTest(hmHDI$Mobile2012, hmHDI$HDI_level, center="mean")
Levene's Test for Homogeneity of Variance (center = "mean")
      Df F value Pr(>F)
group  1  0.0733 0.7875
      64
> |
```

value of the
statistics

p value is greater than 0.05
The variances are equal



Independent Samples t-test

Because of the previously
conducted Levene's test

Independent samples t-test

```
t.test(hmHDI$Mobile2012~hmHDI$HDI_level, var.equal = TRUE)
```

Value of the
statistics

```
> # Independent samples t-test  
> t.test(hmHDI$Mobile2012~hmHDI$HDI_level, var.equal = TRUE)
```

95% confidence
interval for the
mean difference

Two sample t-test

```
data: hmHDI$Mobile2012 by hmHDI$HDI_level  
t = 0.82317, df = 64, p-value = 0.4135  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
-10.55833 25.35756  
sample estimates:  
mean in group high mean in group medium  
120.2650 112.8654
```

Means per group

p value is greater than 0.05
There is no significant difference between these two groups



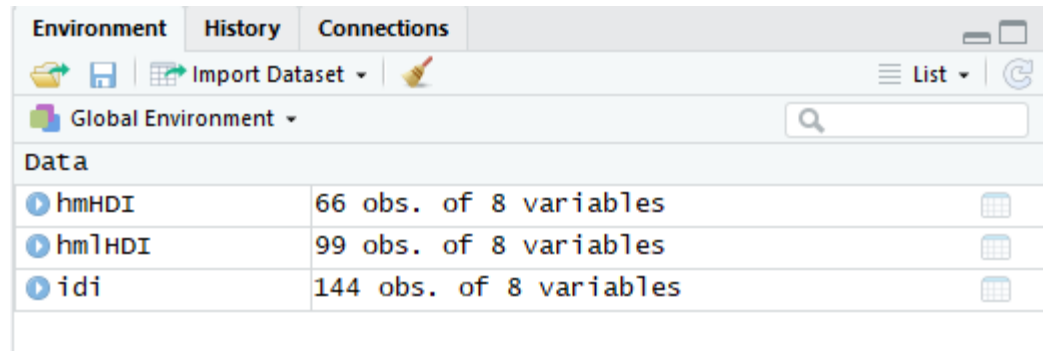
- Two-sample t-test is problematic – compares only two groups
- Analysis of variance (ANOVA) allows us to see if there are differences between more than two means with an **OMNIBUS** test
- The variable must be normally distributed
- Observes variance **within groups** and **between groups**
- H_0 : There is no difference in means
- H_1 : At least one mean differs



- **Example:** Is there a statistically significant difference in the Mobile subscriptions in 2012 between countries with high, medium, and low HDI index?

We will make subset for high/medium/low HDI level

```
hm1HDI <- subset(idi, (HDI_level=="high" | HDI_level=="medium" | HDI_level=="low"))
```



The screenshot shows the R Studio 'Environment' pane. It lists three data objects under the 'Data' section:

Object	Description
hmHDI	66 obs. of 8 variables
hm1HDI	99 obs. of 8 variables
idi	144 obs. of 8 variables



Analysis of variance

```
fit<-aov(hm1HDI$Mobile2012~hm1HDI$HDI_level)  
summary(fit)
```

```
> # Analysis of variance  
> fit<-aov(hm1HDI$Mobile2012~hm1HDI$HDI_level)  
> summary(fit)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
hm1HDI\$HDI_level	2	67901	33951	30.15	6.89e-11	***
Residuals	96	108093	1126			

signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

value of the
statistics

p value is less than 0.05
The means are not equal, at least one mean differs



- ANOVA shows that there is difference between groups, but we do not know which pairs of groups are significantly different from each other.
- Conducting t-test between groups inflates the type I error
- Growing need for an omnibus test to conduct the comparisons
- **Post-hoc tests** are tests of multiple comparison which indicate between which groups there is significant difference



Post-hoc test with no adjustments to p-value

```
#Post-hoc tests for ANOVA (no adjustment for p-values)
pairwise.t.test(hmlHDI$Mobile2012, hmlHDI$HDI_level, p.adj = "none")
```

```
> #Post-hoc tests for ANOVA (no adjustment for p-values)
> pairwise.t.test(hmlHDI$Mobile2012, hmlHDI$HDI_level, p.adj = "none")

      Pairwise comparisons using t tests with pooled SD

data:  hmlHDI$Mobile2012 and hmlHDI$HDI_level

      high      low
low    6.1e-11 -
medium 0.38    9.9e-08

P value adjustment method: none
> |
```



Post-hoc test with adjustments to p-value

```
#Post-hoc tests for ANOVA (Bonferroni adjustment for p-values)
pairwise.t.test(hmlHDI$Mobile2012, hmlHDI$HDI_level, p.adj = "bonf")
```

```
> #Post-hoc tests for ANOVA (Bonferroni adjustment for p-values)
> pairwise.t.test(hmlHDI$Mobile2012, hmlHDI$HDI_level, p.adj = "bonf")

Pairwise comparisons using t tests with pooled SD

data: hmlHDI$Mobile2012 and hmlHDI$HDI_level

      high      low
low  1.8e-10 -
medium 1      3.0e-07

P value adjustment method: bonferroni
> |
```

Bonferroni adjustment is more restrictive



Tukey post-hoc test

#Tukey post-hoc test for ANOVA TukeyHSD(fit)

```
> #Tukey post-hoc test
> TukeyHSD(fit)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = hm1HDI$Mobile2012 ~ hm1HDI$HDI_level)

$`hm1HDI$HDI_level`
              diff            lwr            upr            p adj
low-high      -58.116515  -76.90213  -39.33090  0.0000000
medium-high   -7.399615  -27.52324   12.72401  0.6570873
medium-low    50.716900   29.76935   71.66445  0.0000003
```



Paired samples t-test

- Used to compare the means between two related groups of samples
- The certain variable is measured on the same entities twice
- Example: Results before and after warm up, results before and after therapy, results before and after the introduction of a policy,...
- H_0 : There is no difference in means
- H_1 : There is difference in means



Paired samples t-test

- **Example:** Is there a statistically significant difference in the Mobile subscriptions between 2011 and 2012?

```
# Paired samples t-test
```

```
t.test(idi$Mobile2011, idi$Mobile2012, paired = TRUE, alternative =  
"two.sided")
```

```
> #Paired samples t-test  
> t.test(idi$Mobile2011, idi$Mobile2012, paired = TRUE, alternative = "two.sided")
```

Paired t-test

```
data: idi$Mobile2011 and idi$Mobile2012  
t = -7.3619, df = 143, p-value = 1.311e-11  
alternative hypothesis: true difference in means is not equal to 0  
95 percent confidence interval:  
 -6.834067 -3.940933  
sample estimates:  
mean of the differences  
      -5.3875
```

If the mean difference is
less than 0 – than
 $m_{2011} < m_{2012}$



- Nonparametric equivalent of a t-test for two independent samples
- We use it when the data is not normally distributed or is ordinal (e.g. Likert scale)
- Very efficient and commonly used
- H_0 : The distributions are equal
- H_1 : The distributions differ



- **Example:** Is there a statistically significant difference in the Fixed subscriptions in 2012 between countries with high and medium HDI index?

```
# We will use the already made subset for high/medium HDI level  
hmHDI <- subset(idi, (HDI_level=="high" | HDI_level=="medium"))
```

Environment		History	Connections
   Import Dataset ▾ 			
 Global Environment ▾			
Data			
 hmHDI	66 obs. of 8 variables		
 idi	144 obs. of 8 variables		
 lottery	285 obs. of 9 variables		
values			



```
# Mann-whitney test  
install.packages("coin")  
library(coin)  
wilcox_test(hmHDI$Fixed2012~hmHDI$HDI_level)
```

```
> wilcox_test(hmHDI$Fixed2012~hmHDI$HDI_level)
```

Asymptotic wilcoxon-Mann-whitney Test

```
data: hmHDI$Fixed2012 by hmHDI$HDI_level (high, medium)  
Z = 4.1407, p-value = 3.463e-05  
alternative hypothesis: true mu is not equal to 0
```



- Non-parametric alternative to one-way ANOVA
- "One-way ANOVA on ranks"
- Extension of the MW test
- H_0 : All of the k population distribution functions are identical
- H_1 : At least one of the populations differs



- **Example:** Is there a statistically significant difference in the Fixed subscriptions in 2012 between countries with high, medium and low HDI index?

```
#Kruskal-wallis test
```

```
kruskal.test(Fixed2012 ~ HDI_level, data = hm1HDI)
```

```
> #Kruskal wallis test
> kruskal.test(Fixed2012 ~ HDI_level, data = hm1HDI)

    Kruskal-wallis rank sum test

data:  Fixed2012 by HDI_level
Kruskal-wallis chi-squared = 69.269, df = 2, p-value = 9.086e-16

> |
```



- Way I – Pairwise test

```
#Kruskal-wallis post-hoc test  
pairwise.wilcox.test(hm1HDI$Fixed2012, hm1HDI$HDI_level,  
p.adjust.method = "bonf")
```

Pairwise comparisons using wilcoxon rank sum test

data: hm1HDI\$Fixed2012 and hm1HDI\$HDI_level

	high	low
low	8.0e-13	-
medium	0.00011	1.2e-08

P value adjustment method: bonferroni

warning messages:

```
1: In wilcox.test.default(xi, xj, paired = paired, ...) :  
  cannot compute exact p-value with ties  
2: In wilcox.test.default(xi, xj, paired = paired, ...) :  
  cannot compute exact p-value with ties  
3: In wilcox.test.default(xi, xj, paired = paired, ...) :  
  cannot compute exact p-value with ties
```

The test is not
precise when there
are ties



- Way II – Dunn's test

#Dunn's test

```
install.packages("dunn.test")
library(dunn.test)
dunn.test(hm1HDI$Fixed2012, hm1HDI$HDI_level, method =
"bonferroni", kw=TRUE)
```

```
data: x and group
Kruskal-wallis chi-squared = 69.2693, df = 2, p-value = 0
```

Comparison of x by group
(Bonferroni)

Col Mean- Row Mean	high	low
low	8.303913 0.0000*	
medium	2.995971 0.0041*	-4.568757 0.0000*

```
alpha = 0.05
Reject Ho if p <= alpha/2
```



Wilcoxon Signed-Rank Test

- Nonparametric alternative of the paired samples t-test

#Wilcoxon Signed-Rank Test

```
wilcox.test(hmlHDI$Fixed2011, hmlHDI$Fixed2012, paired=TRUE)
```

```
> #wilcoxon Signed-Rank Test
> wilcox.test(hmlHDI$Fixed2011, hmlHDI$Fixed2012, paired=TRUE)

      wilcoxon signed rank test with continuity correction

data:  hmlHDI$Fixed2011 and hmlHDI$Fixed2012
V = 1949, p-value = 0.04209
alternative hypothesis: true location shift is not equal to 0

> |
```



Contingency table

- A contingency table is a special type of frequency distribution table, where two variables are shown simultaneously.
- Contingency table test is used to examine the relationship between two nominal or ordinal variables.
- H_0 : X and Y are independent
- H_1 : X and Y are dependent



Contingency table

Crosstabs

```
install.packages("gmodels")  
install.packages("foreign")  
library("gmodels")  
library(foreign)  
lottery<- read.spss(file.choose(), to.data.frame = TRUE)  
Crosstabs(lottery$LotterySpending, lottery$LotteryPlaying,  
prop.r=F, prop.c=F, prop.t=F, prop.chisq=F, chisq=T)
```

The percentages
per row and column
are set not to be
calculated (Set to
False - F)

The total percentage
of cases per cell is
set not to be
calculated (Set to
False - F)

The difference between the
measured and theoretical
frequencies is set not to be
calculated (Set to False - F)



Contingency table

Crosstabs

```
CrossTable(lottery$LotterySpending, lottery$LotteryPlaying,
prop.r=F, prop.c=F, prop.t=F, prop.chisq=F, chisq=T)
```

Total Observations in Table: 285

lottery\$LotterySpending	lottery\$LotteryPlaying Rarely	Several times a month	Several times a week	Row Total
over 500 dinars	5	11	21	37
up to 500 dinars	227	18	3	248
Column Total	232	29	24	285

Statistics for All Table Factors

Pearson's Chi-squared test

Chi^2 = 158.0211 d.f. = 2 p = 4.854669e-35

value of the
statistics

p value is greater than 0.05
The two variables are independent



Contingency table

Crosstabs

```
CrossTable(lottery$LotterySpending, lottery$LotteryPlaying,
prop.r=T, prop.c=F, prop.t=F, prop.chisq=F, chisq=T)
```

Total observations in Table: 285

lottery\$LotterySpending	lottery\$LotteryPlaying Rarely	Several times a month	Several times a week	Row Total
over 500 dinars	5 0.135	11 0.297	21 0.568	37 0.130
up to 500 dinars	227 0.915	18 0.073	3 0.012	248 0.870
column Total	232	29	24	285

Statistics for All Table Factors

Pearson's Chi-squared test

chi^2 = 158.0211 d.f. = 2 p = 4.854669e-35



Contingency table

Crosstabs

```
CrossTable(lottery$LotterySpending, lottery$LotteryPlaying,
prop.r=F, prop.c=T, prop.t=F, prop.chisq=F, chisq=T)
```

lottery\$LotterySpending	lottery\$LotteryPlaying			
	Rarely	Several times a month	Several times a week	Row Total
over 500 dinars	5 0.022	11 0.379	21 0.875	37
up to 500 dinars	227 0.978	18 0.621	3 0.125	248
Column Total	232 0.814	29 0.102	24 0.084	285

Statistics for All Table Factors

Pearson's Chi-squared test

Chi^2 = 158.0211 d.f. = 2 p = 4.854669e-35



Contingency table

Crosstabs

```
CrossTable(lottery$LotterySpending, lottery$LotteryPlaying,
prop.r=F, prop.c=F, prop.t=T, prop.chisq=F, chisq=T)
```

lottery\$LotterySpending	lottery\$LotteryPlaying Rarely	Several times a month	Several times a week	Row Total
over 500 dinars	5 0.018	11 0.039	21 0.074	37
up to 500 dinars	227 0.796	18 0.063	3 0.011	248
Column Total	232	29	24	285

Statistics for All Table Factors

Pearson's Chi-squared test

Chi^2 = 158.0211 d.f. = 2 p = 4.854669e-35



Contingency table

Crosstabs

```
CrossTable(lottery$LotterySpending, lottery$LotteryPlaying,
prop.r=T, prop.c=T, prop.t=T, prop.chisq=F, chisq=T)
```

lottery\$LotterySpending	lottery\$LotteryPlaying			Row Total
	Rarely	Several times a month	Several times a week	
over 500 dinars	5 0.135 0.022 0.018	11 0.297 0.379 0.039	21 0.568 0.875 0.074	37 0.130
up to 500 dinars	227 0.915 0.978 0.796	18 0.073 0.621 0.063	3 0.012 0.125 0.011	248 0.870
Column Total	232 0.814	29 0.102	24 0.084	285

Statistics for All Table Factors

Pearson's Chi-squared test

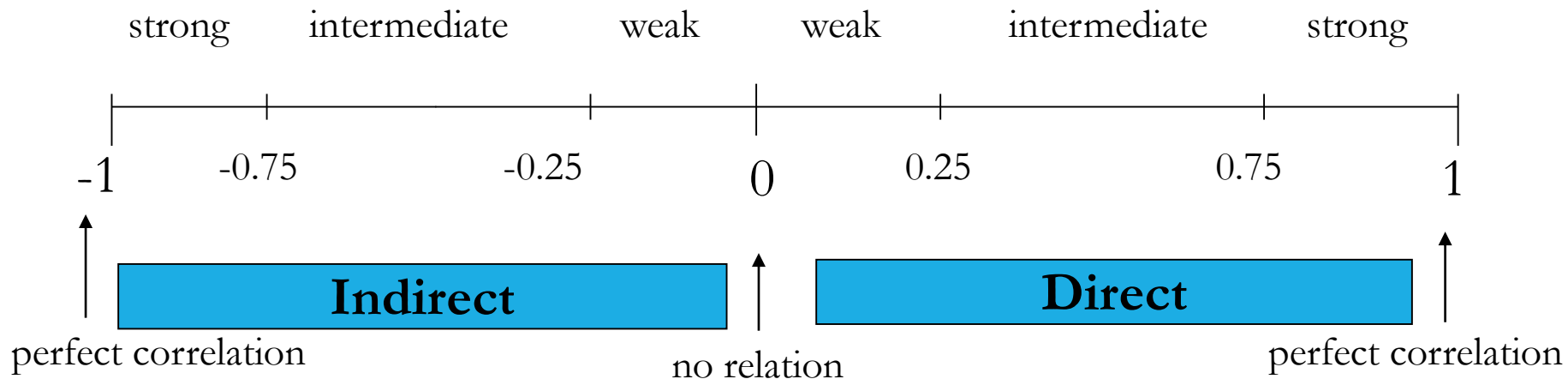
Chi^2 = 158.0211 d.f. = 2 p = 4.854669e-35



- It measures the nature and strength of the relationship between two quantitative variables
- The **sign** of r denotes the nature of association
- The **value** of r denotes the strength of association
- The value of r ranges between (-1) and (+1)



- The value of r denotes the strength of the association as illustrated by the following diagram.



- Pearson's correlation coefficient
 - The one we calculated by hand on classes
 - Both variables should be normally distributed
- Spearman's correlation coefficient
 - Non-parametric alternative to Pearson's correlation coefficient
 - Both variables are quantitative but not normally distributed
 - Both variables are quantitative but are ordinal
 - One variable is quantitative and the other is qualitative ordinal



- **Example:** What is the relationship between Mobile subscriptions in 2011 and in 2012?

```
# Method can be pearson, spearman or kendall  
cor(idi$Mobile2011, idi$Mobile2012, method="pearson")
```

```
> # Correlations  
> # Method can be pearson, spearman or kendall  
> cor(idi$Mobile2011, idi$Mobile2012, method="pearson")  
[1] 0.9758
```

Value of Pearson's
correlation coefficient



- Previous analysis does not produce test of significance
- H_0 : The correlation coefficient is equal to 0
- H_1 : The correlation coefficient is different from 0
- Therefore, another function should be called



```
# the cor.test function to test a single correlation coefficient  
cor.test(idi$Mobile2011, idi$Mobile2012, method="pearson")
```

value of the
statistics

```
> cor.test(idi$Mobile2011, idi$Mobile2012, method="pearson")
```

Pearson's product-moment correlation

data: idi\$Mobile2011 and idi\$Mobile2012

t = 53.177, df = 142, p-value < 2.2e-16

alternative hypothesis: true correlation is not equal to 0

95 percent confidence interval:

0.9664932 0.9825447

sample estimates:

cor

0.9758

Pearson's
correlation
coefficient

95% confidence
interval for the r

p value is less than 0.05
r is different from 0

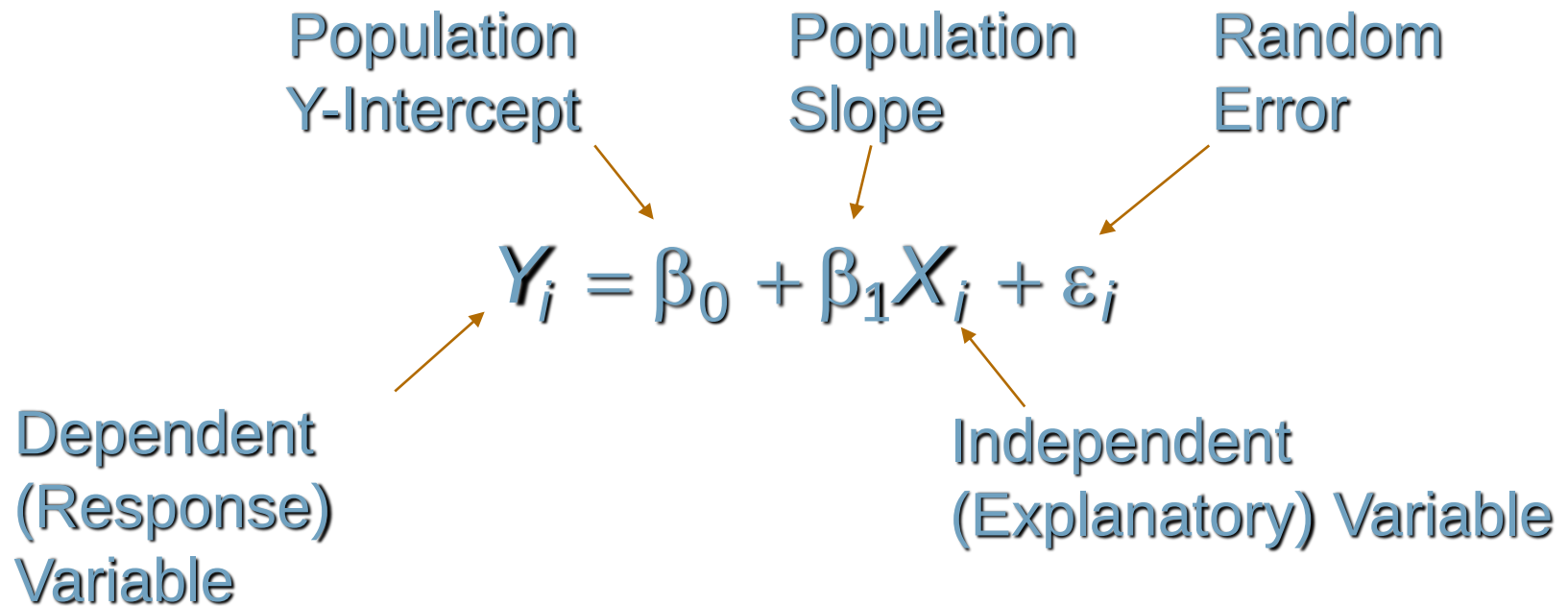


Linear regression

- Uses variable (x) or variables ($x_1, x_2, \dots, x_k$) to predict some dependent variable (y)
- The value of the dependent variable (e.g. Mobile-cellular subscriptions) can be predicted based on its relationship to the independent variables used in the analysis (Fixed-telephones subscriptions).



Simple Linear regression - Population



Population Y-Intercept

Population Slope

Random Error

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

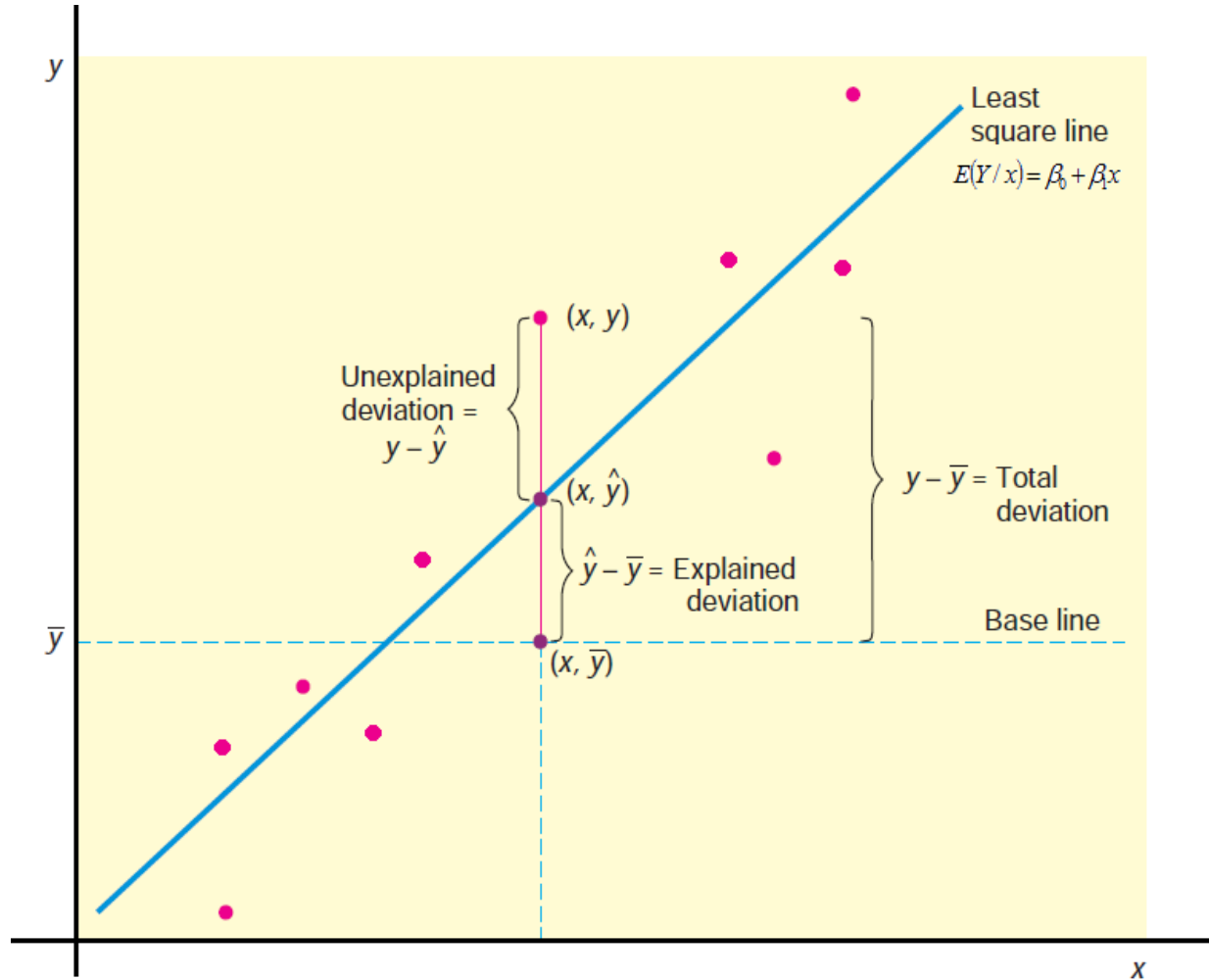
Dependent (Response) Variable

Independent (Explanatory) Variable

The diagram illustrates the Simple Linear Regression equation for the population. The equation is $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$. Arrows point from descriptive labels to the corresponding terms in the equation: 'Population Y-Intercept' points to β_0 , 'Population Slope' points to β_1 , 'Random Error' points to ε_i , 'Dependent (Response) Variable' points to Y_i , and 'Independent (Explanatory) Variable' points to X_i .



Simple Linear regression



Simple linear regression - Sample

- Coefficients β_0 and β_1 are unknown and should be estimated
- Coefficient β_0 represents the Intercept – segment of the Y-axis
- Coefficient β_1 represents the Slope – indicates for how many units will Y change if X changes for 1 unit
- The estimated model is:

$$\hat{Y}_i = b_0 + b_1 X_i, \quad i = 1, 2, \dots, n$$



1. Test on the significance of the coefficient β_1
 - H_0 : The coefficient β_1 is equal to 0
 - H_1 : The coefficient β_1 is different from 0

2. Test on the significance of the whole model
 - H_0 : The model is not statistically significant
 - H_1 : The model is statistically significant



Coefficient of determination

- It is interpreted as the proportion of the variance in the dependent variable that is predictable from the independent variable
- An R^2 of 0 means that the dependent variable cannot be predicted from the independent variable
- An R^2 of 1 means the dependent variable can be predicted without error from the independent variable
- An R^2 of 0.10 means that 10 percent of the variance in Y is explained by X



- A scatter plot displays the nature of the relationship between two variables
- **Before** performing a regression analysis, it is recommended to create a scatter plot to determine if there is a linear relationship between the variables.
- Rectangular coordinate
- X Axis – Independent variable
- Y Axis – Dependent variable



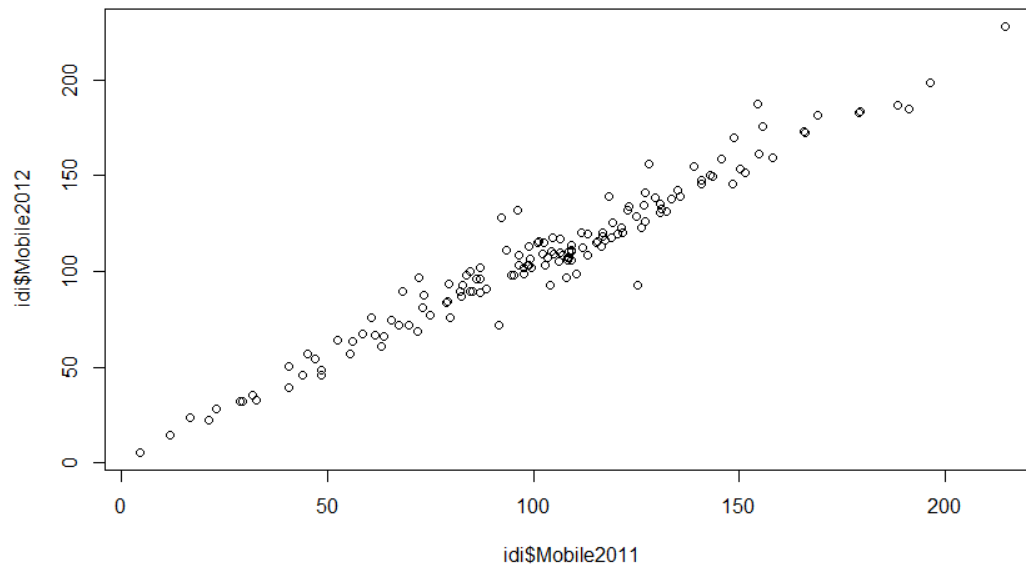
Examples:

1. Using scatter plot describe the relationship between Mobile-cellular subscriptions in 2011 and 2012
2. Using linear regression model the Mobile-cellular subscriptions in 2012 using Mobile-cellular subscriptions in 2011. Is the model statistically significant?
3. Which proportion of Mobile-cellular subscriptions in 2012 can be explained using Mobile-cellular subscriptions in 2011?



Scatter plot – No labels

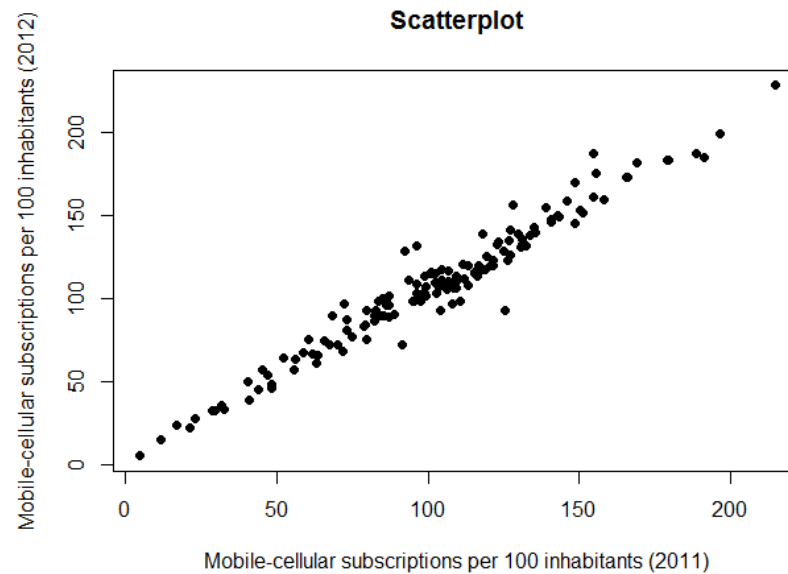
```
# Simple Scatterplot without labels  
plot(idi$Mobile2011, idi$Mobile2012)
```



Scatter plot

Simple Scatterplot

```
plot(idi$Mobile2011, idi$Mobile2012, main="Scatterplot",  
     xlab="Mobile-cellular subscriptions per 100 inhabitants (2011)",  
     ylab="Mobile-cellular subscriptions per 100 inhabitants (2012)",  
     pch=19)
```



Examples:

1. Using scatter plot describe the relationship between Mobile-cellular subscriptions in 2011 and 2012
 - The relationship is **strong, direct, and linear**



Linear Regression

```
fit_s <- lm(idi$Mobile2012 ~ idi$Mobile2011, data=idi)
```

```
# Show results
```

```
summary(fit_s)
```

```
Console ~/
> fit <- lm(Mobile2012 ~ Mobile2011, data=IDI)
> summary(fit)

Call:
lm(formula = Mobile2012 ~ Mobile2011, data = IDI)

Residuals:
    Min       1Q   Median       3Q      Max
-38.002  -4.464  -0.997   3.709  30.719

Coefficients:
(Intercept)  5.32392  2.06119  2.583  0.0108 *
Mobile2011   1.00062  0.01882 53.177 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.813 on 142 degrees of freedom
Multiple R-squared:  0.9522, Adjusted R-squared:  0.9518
F-statistic: 2828 on 1 and 142 DF, p-value: < 2.2e-16

> |
```

b_0 : Intercept

b_1 : Slope

value of the
statistics for b_1

p value of the
statistics for the b_1



Linear Regression

```
fit_s <- lm(idi$Mobile2012 ~ idi$Mobile2011, data=idi)
```

Show results

```
summary(fit_s)
```

```
Console ~/
> fit <- lm(Mobile2012 ~ Mobile2011, data=IDI)
> summary(fit)

Call:
lm(formula = Mobile2012 ~ Mobile2011, data = IDI)

Residuals:
    Min       1Q   Median       3Q      Max
-38.002  -4.464  -0.997   3.709  30.719

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.32392    2.06119   2.583   0.0108 *
Mobile2011    1.00062    0.01882  53.177  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.813 on 142 degrees of freedom
Multiple R-squared: 0.9522, Adjusted R-squared: 0.9518
F-statistic: 2828 on 1 and 142 DF, p-value: < 2.2e-16
```

Proportion of variation that is explained by the model.

value of the statistics related to the significance of the whole model

p value of the statistics for the whole model



Examples:

2. Using linear regression model the Mobile-cellular subscriptions in 2012 using Mobile-cellular subscriptions in 2011. Is the model statistically significant?

In this case, the values of $\mathbf{b}_0$ and $\mathbf{b}_1$ are as follows: $\mathbf{b}_0 = 1.001$, $\mathbf{b}_1 = 5.324$

$$\hat{Y}_i = 1.001 + 5.324X_i$$

Both the coefficient and the model are statistically significant as p values are less than 0.05



Examples:

3. Which proportion of Mobile-cellular subscriptions in 2012 can be explained using Mobile-cellular subscriptions in 2011?

The coefficient of determination is 0.9522, therefore 95.22% of variability of Mobile-cellular subscriptions in 2012 can be explained with Mobile-cellular subscriptions in 2011



Multiple regression

- **Multiple regression** estimates the coefficients of the linear equation when there is more than one independent variable that best predicts the value of the dependent variable. For example, a salesperson's total annual sales (the dependent variable) can be predicted based on independent variables such as age, education, and years of experience. The linear equation for multiple regression is as follows:

$$Y = b_1X_1 + b_2X_2 + b_0$$

- **Predicting Values of Dependent Variables**
- Mobile-cellular subscriptions for the year of 2012 (the dependent variable) can be predicted from more than one independent variable (such as Fixed-telephones subscriptions for the year of 2011 and Mobile-cellular subscriptions for the year of 2011) by using multiple regression analysis.



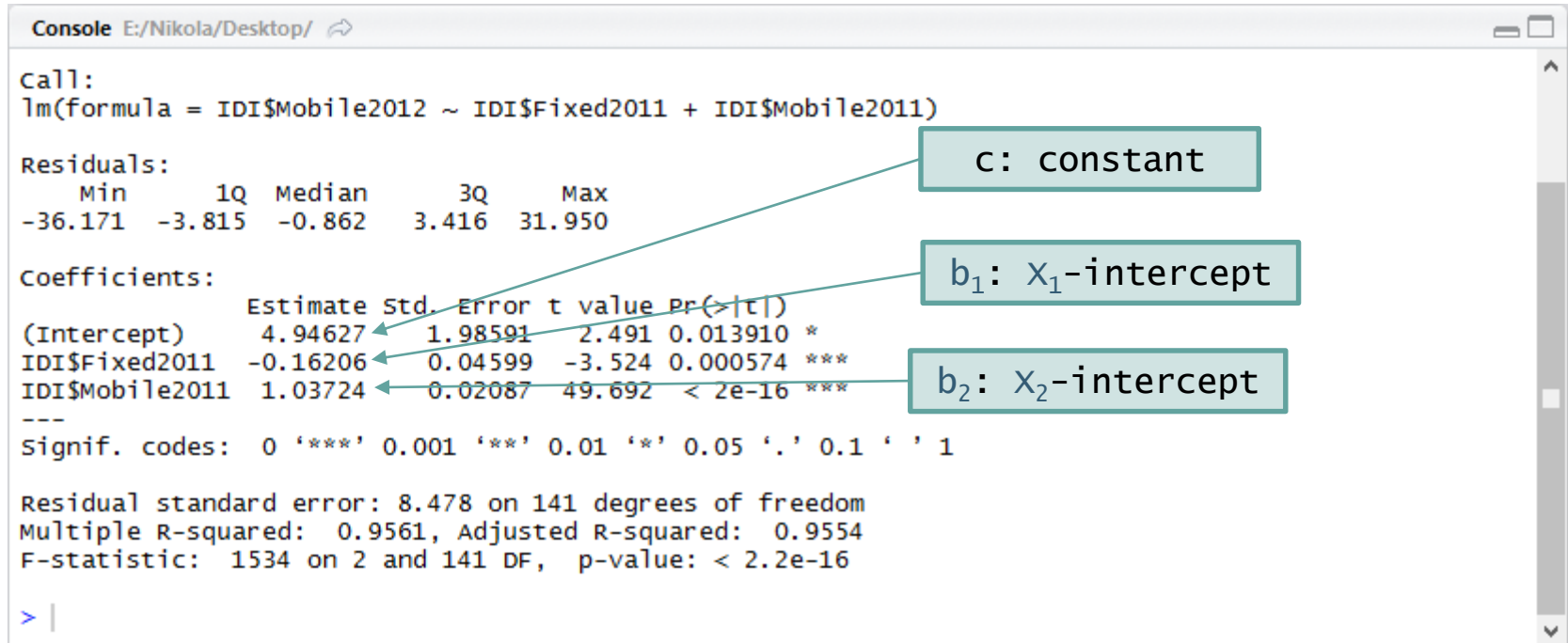
Multiple regression

Multiple Regression

```
fit_m <- lm(idi$Mobile2012 ~ idi$Fixed2011 + idi$Mobile2011)
```

show results

```
summary(fit_m)
```



```
Console E:/Nikola/Desktop/ ↗  
Call:  
lm(formula = IDI$Mobile2012 ~ IDI$Fixed2011 + IDI$Mobile2011)  
  
Residuals:  
    Min       1Q   Median       3Q      Max   
-36.171  -3.815  -0.862   3.416  31.950   
  
Coefficients:  
            Estimate Std. Error t value Pr(>|t|)      
(Intercept)  4.94627    1.98591   2.491 0.013910 *      
IDI$Fixed2011 -0.16206    0.04599  -3.524 0.000574 ***     
IDI$Mobile2011 1.03724    0.02087  49.692 < 2e-16 ***     
---  
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
Residual standard error: 8.478 on 141 degrees of freedom  
Multiple R-squared:  0.9561, Adjusted R-squared:  0.9554  
F-statistic: 1534 on 2 and 141 DF,  p-value: < 2.2e-16  
  
> |
```

c: constant

b_1 : X_1 -intercept

b_2 : X_2 -intercept



Multiple regression

Multiple Regression

```
fit_m <- lm(idi$Mobile2012 ~ idi$Fixed2011 + idi$Mobile2011)
```

show results

```
summary(fit_m)
```

```
Console E:/Nikola/Desktop/ ↗

Call:
lm(formula = IDI$Mobile2012 ~ IDI$Fixed2011 + IDI$Mobile2011)

Residuals:
    Min       1Q   Median       3Q      Max
-36.171  -3.815  -0.862   3.416  31.950

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.94627    1.98591   2.491  0.013910 *
IDI$Fixed2011 -0.16206    0.04599  -3.524  0.000574 ***
IDI$Mobile2011 1.03724    0.02087  49.692 < 2e-16 ***
---
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Residual standard error: 8.478 on 141 degrees of freedom
Multiple R-squared:  0.9561, Adjusted R-squared:  0.9554
F-statistic: 1534 on 2 and 141 DF,  p-value: < 2.2e-16

> |
```

Statistics and p value for b_1

Statistics and p value for b_2



Multiple regression

Multiple Regression

```
fit_m <- lm(idi$Mobile2012 ~ idi$Fixed2011 + idi$Mobile2011)
```

show results

```
summary(fit_m)
```

```
Console E:/Nikola/Desktop/ ↗

Call:
lm(formula = IDI$Mobile2012 ~ IDI$Fixed2011 + IDI$Mobile2011)

Residuals:
    Min       1Q   Median       3Q      Max
-36.171  -3.815  -0.862   3.416  31.950

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   4.94627    1.98591   2.491 0.013910 *
IDI$Fixed2011 -0.16206    0.04599  -3.524 0.000574 ***
IDI$Mobile2011 1.03724    0.02087  49.692 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.478 on 141 degrees of freedom
Multiple R-squared:  0.9561, Adjusted R-squared:  0.9554
F-statistic: 1534 on 2 and 141 DF, p-value: < 2.2e-16

> |
```

Statistics and p
value for the
entire model



Predicting Mobile-cellular subscriptions for the year of 2013 with Multiple Regression Model

- To predict Mobile-cellular subscriptions for the year of 2013 for each country, substitute the values of a, b, and c in the following linear equation:

$$Y = b_1X_1 + b_2X_2 + b_0$$

$$\text{Mobile2012} = -0.162 * \text{Fixed2011} + 1.037 * \text{Mobile2011} + 4.946$$



Introduction to Data Analysis